SINA: A Leakage-Certified Benchmark for Disease-Associated Gene Recovery on OptimusKG

SINA: A Leakage-Certified, Pathway-Aware Benchmark for Disease-Associated Gene Recovery on OptimusKG

Reza Alipour et al. — SINA Platform — updated 2026-10-06 (M2b frozen 2026-08-28, M2c completion track 2026-09-05→06, external head-to-head + leakage-controlled re-grade 2026-09-11→25, M4B tabular + M5B functional tracks 2026-09-12→26)
Branch `main` — `reports/SINA_PAPER.md` is the single reproducible record (historical pins: M2b sections at `53f24ca`, M2c sections at `ab23259`; §2.10-2.11 / §3.8-3.10 supersede the per-locus framing of the pinned versions)
Code: `https://github.com/Abolfazl-Alipour/sina-platform.git` (private, `main`) — **Data:** `OptimusKG` DOI `10.7910/DVN/IYNGEV` (code `MIT`, data per-source, some `NC` per-source), `PrimeKG` DOI `10.7910/DVN/IXA7BM` (`CC BY-NC-SA 4.0`), `MONDO` `2026-07-24` `53,086,111` bytes `sha256 fce4bccd...` (`CC-BY 4.0`) — **EBS persist v2:** `sina_full_backup_v2_2026-08-27.tar.gz` `17GB→5.5GB` (`~/persist` + `~/sina-platform` + `~/miniconda3/envs/{sina,biopathnet}` + `cuda-11.8-overlay`, `RECOVERY.md` on EBS, zero-context restore via `FRESH_INSTANCE_HANDOFF.md`)
License: Code `MIT`, Data per-source (see `configs/m1_trackc_sources.yaml` + `configs/m2_trackd_relations.yaml`)

Abstract

We present SINA, a pre-registered, leakage-controlled benchmark for disease-associated gene recovery with graph neural networks, built on a frozen 1,431,945-edge background graph derived from OptimusKG (190,531 nodes / 21.8M edges) with 191,953 disease–gene supervision edges over 678 diseases. A 6-layer BioPathNet/NBFNet is evaluated under strict leave-one-disease-out (LODO) zero-shot conditions — every association of the held-out disease withheld — against same-graph degree, personalized-PageRank, and degree-matched nulls, with degree-preserving shuffle controls. **The primary result is a certified negative:** across 13 primary diseases the model reaches mean MRR 0.00775 (3 seeds, 130× random but not significant, Wilcoxon p=0.34); 16 secondary units agree (p=0.97); shuffled controls collapse 58–89%, proving the exam leakage-free. A pathway-level funnel fares no better (Recall@10 0.0; its anchored variant ties a topology null). A complementary seen-disease completion track (Parkinson's, Alzheimer's) plateaus at test MRR 0.019/0.011 — above the degree floor but tied with a supervision-only counting null on Hits@10, invariant to loss, width, batch, and edge-set ablations; prospective prediction of post-2020 GWAS associations collapses for model and null alike, and the functional-gold final test correctly refused to run at n=2 vs the pre-registered 15. An external head-to-head against Open Targets L2G shows the model's per-locus strength (top-1 0.904/0.954) exists only for genes present in its training supervision: on unseen genes it loses (PD 0.300 vs 0.600, n=10) or ties (AD 0.161 vs 0.226, n=31) the external baseline, below chance-draw means. Two CPU controls bound the task itself: a leakage-controlled tabular model ties Open Targets on covered loci (+0.006 [−0.039,+0.053], n=491) while significantly beating nearest-gene distance (+0.057 [+0.014,+0.104]), and five added functional-evidence tracks all lose to Open Targets (best PR AUC 0.300 vs 0.477). A replication of the shipped FLAMES benchmark finds its reported win over Open Targets is not significant on top-1. We release the complete harness — frozen 29-disease roster, hash-pinned manifests, negative controls, decision records, and a clean-machine reproduction log — as a falsification instrument. On this supervision, the ceiling is the labels, not the model.

Table of Contents

1. Introduction + Contributions
2. Methods — Exact Reproduction (Data, BRG correction, LODO roster, Exports, Encoding, Model, Training, Evaluation, Pathway Funnel, Reproducibility, Completion track, External head-to-head, Tabular/functional controls)
3. Results (Gate0, Step0b, Wave1, Three-seed, Wave2, Shuffle, Pathway + Anchored, Completion PD/AD, Head-to-head vs OT

L2G, M4B tabular + FLAMES replication, M5B functional arms)
4. Discussion — what survives hostile review
5. Limitations
6. Data and Code Availability — complete file inventory
7. How to reproduce — copy-paste commands
8. References
9. Appendices (A: Registry, B: Config hashes, C: Per-unit details, D: Encoding worked example)
10. Peer-review brief — project state, open questions, how to give feedback

---

1. Introduction

Biomedical knowledge graphs promise to prioritize candidate genes for unseen diseases, but most benchmarks leak through random-edge splits, popularity bias, or phenotype-adjacent shortcuts. SINA was designed as a **falsification-first** platform: every design choice is pre-registered in `CHANGES/` and guarded by hash-pinned manifests, same-graph nulls, and degree-preserving shuffles. The platform is not a therapeutic engine — it is a hypothesis-triage funnel whose outputs feed `3D organoid CRISPR` validation (`docs/organoid_handoff_spec_stub.md`).

What SINA is not: a copy of BioPathNet's paper. BioPathNet (`Nature BME 10.1038/s41551-025-01598-z`, code `b051a90 MIT` at `third_party/biopathnet`) evaluated `drug→disease`, `gene→function` (`MRR 0.547 Hits@10 72.3%`), `synthetic lethality` (`MRR 0.31→0.36 Hits@10 44→50%`), `lncRNA→gene` (`MRR 0.19`) on `PrimeKG` (`8.1M` edges) with **random edge splits** (hide `10%` of links, `90%` of that disease stays in train). SINA evaluates `disease→gene` **zero-shot LODO** on `OptimusKG 1.43M`: hide the entire disease (`730` Werner genes) and predict from other diseases. Harder (`rank 128` vs `rank 2`), leakage-free.

Contributions: (i) the first `disease→gene` LODO benchmark on `OptimusKG` with a frozen `29 =13+5+11` roster (`0=5%`); (ii) a measured cost model (`1.43M` graph, `97.8 min/pass`); (iii) a `12`-arm `KEEP/1-pass` ablation; (iv) `13/13` primary `1-seed + 3-seed BCa` evaluation; (v) `6/6` shuffle collapse proof; (vi) a pathway-level funnel that pools `1,000` gradient-weighted traces into `10` pathways and is gated on the same shuffle collapse, plus its anchored-vs-topology-null control that reveals the exploratory table as artifact; (vii) a disaster-recovery-hardened EBS persist (`v2 5.5GB`) with zero-context `FRESH_INSTANCE_HANDOFF.md`; (viii) honest negative result with full file inventory; (ix) a completion track (M2c, §2.9/§3.7) that re-tests the stack at full-epoch scale on two diseases under a frozen rank-loss criterion, then closes the discovery question with prospective and functional-gold probes; (x) an external head-to-head against Open Targets L2G under one protocol with a leakage-controlled known-gene split (§2.10/§3.8) that shows the per-locus recovery headline

is train-positive memorization, not unseen-gene signal; (xi) a leakage-controlled tabular control (M4B, §2.11/§3.9) that ties OT L2G on covered loci and beats nearest-gene distance significantly — plus a FLAMES replication that overturns its published OT comparison on top-1; (xii) five pre-registered functional-evidence tracks (M5B, §2.11/§3.10) that all lose to OT, bounding what public functional data adds on this task.

---

2. Methods — Exact Reproduction

2.1 Data sources and BRG correction — with the 295k bug

- **OptimusKG** `v2.0` `doi:10.7910/DVN/IYNGEV` (Harvard Zitnik Lab, `optimuskg==1.0.0`, `Polars` Parquet). Canonical counts `190,531` nodes / `21,813,816` edges / `27` relations (`DIS-GEN 9,734,774` incl. `year_initial/year_final`, `ANA-GEN 8,787,955` Bgee, `GEN-GEN 327,924` `PrimeKG`-derived). **Per-source licenses, some `NC`** — `Track-D` research-only, `Track-C` `MIT` skeleton only. Download `37` stratified files verified via `data/raw/REGISTRY.json` (`SHA-256` + `size` per asset).
- **PrimeKG** `v2.1` `doi:10.7910/DVN/IXA7BM` (`129,312` nodes / `8,100,498` edges) — reproduction anchor only. `alzheimers_42` split (`42` Alzheimer diseases) is `8.1M` minus held-out, `4` full epochs `batch 16` (`third_party/biopathnet/config/primekg/alzheimers.yaml`).
- **MONDO** `2026-07-24` `http://purl.obolibrary.org/obo/mondo.obo` `53,086,111` bytes `sha256 fce4bccd...` (`CC-BY 4.0`) — `EFO/DOID/Orphanet→MONDO` crosswalk (`42.3%` coverage on `Track-D`).
- **BioPathNet paper claims (for context, not our numbers):** `gene function MRR 0.547 Hits@10 72.3%`, `synthetic lethality MRR 0.31→0.36 Hits@10 44→50%`, `lncRNA MRR 0.19`, `Alzheimer drug 7/8 known in top 14 (4/7 FDA)` — all **random edge split**, not LODO.
- **BRG correction — the $21 H200 was on broken graph:** `scripts/m2_build_trimmed_taskg.py` had `src_id.is_in(sup_genes)` on `annotation→gene` edges (`src=GO/BPO/CCO/MFN/PWY/PHE` term) → kept `0/1.1M` annotation edges → `train1 ≈295k` (only `GEN-GEN`) → fixed `dst_id.is_in(sup_genes)` (`34c1d32` `2026-08-08`) → **`train1 = 1,431,945`** (`1,420,095` after `DIS-PHE DROP`, `1,431,945` with `KEEP`). Old `142/144GB @bs300` on `295k` is invalid and not comparable. **Focused trimmed cluster** for feasibility: `678` diseases / `191,953` `DIS-GEN` / `17,448` genes / `1,270` drugs — `BRG 1,431,945` (`PHE-GEN 768,844 · GEN-GEN 275,220 · BPO 147,477 · CCO 94,150 · MFN 82,362 · PWY 43,754 · DIS-PHE 11,850 · DRG-GEN 5,310 · DRG-DIS 1,853 · DIS-DIS 1,125`). `ANA-GEN 8.79M` dropped from training (tissue co-occurrence hub) and added back post-hoc as annotation only.

2.2 LODO task and roster — frozen CHANGES/0020 2026-08-08

- **Task:** disease --ASSOCIATED_WITH→ ? over genes only. Supervised positives/negatives are DIS-GEN only; train1 is message-passing only (never labels, never loss).
- **LODO vs random:** Random hides 10% of edges for same disease (seen disease, easy). LODO hides 100% of disease cluster (unseen disease, hard). Real clinic for known Alzheimer = random-like; for new rare disease = LODO-like. SINA picks LODO to be leakage-free.
- **Roster frozen CHANGES/0020 (D1 3-panel + D2 Kufor-Rakeb+PSP-parkinsonism sibling-strip):** 29 =13 primary +5 secondary-evidence +11 secondary-low-context (0=5% ceiling, A1-A8 validated, 2/29 degenerate, 29/29 ≤2-hop handled, exact Wilcoxon 2^13 enumerated, 38 pairs, --dry-run EXIT 0). lean-10 CHANGES/0023 superseded 2026-08-12 (p=0.125 destroyed 4-cluster inference). Family configs/disease_families.yaml OWNER-APPROVED 2026-07-25.
- **Full roster table:** see reports/m2b/lodo_exports_keep.md and data/processed/m2/lodo_exports/manifest.parquet.

2.3 Exports — per-unit BioPathNet-format

- **Location:** data/processed/m2/lodo_exports/<unit>{_keep_disphe}/ where : → __ (e.g., mondo__0010196). Each contains train1.txt (1,431,945 KEEP, identical across units, no DIS-GEN), train2.txt (supervision minus held-out cluster + ancestors 9-45 + siblings 5-185 via strip_list.json), valid 15% / test 85% by evidence-group ID 1337 (reciprocal twins co-assigned, disjointness asserted; train2∩train1=0, valid/test∩train1=0), entity_names.txt, entity_types.txt, strip_list.json. 678→191,953 pool, per-unit train2 52-730 (primaries, see table in §3.3) + valid 7-126 + test 52-730.

unit	name	tier	train2	valid	test	anc	sib
0007669	renal cysts and diabetes	primary	81,624	49	258	21	24
0008199	late-onset Parkinson	primary	39,274	9	57	9	108
0008310	HGPS	primary	58,021	73	394	28	28
0010196	Werner	primary	59,454	126	730	28	36
…	…	…	…	…	…	…	…

unit	name	tier	train2	valid	test	anc	sib
(full 29 in lodo_exports_keep.md)							

- **Build scripts:** scripts/m2_build_trimmed_taskg.py (BRG), scripts/m2_build_lodo_exports.py (LODO), scripts/m2_build_crosswalk.py (MONDO), all REGISTRY.json hash-verified, manifest.json emitted.

2.4 Encoding — nodes and edges, with jargon explained

Node encoding: nn.Embedding(double_relation, 32) in third_party/biopathnet/biopathnet/model.py:52 — learnable 32-dim vectors. At t=0, boundary = zeros(num_nodes, batch, 32) then boundary[h] = query(ASSOCIATED_WITH) where query = Embedding [model.py:114-120]. So **source disease (e.g., Werner) = query vector**, all other 17,447 nodes = zero. Not random per node — indicator initialization.

Edge/Relation encoding: self.relation = nn.Embedding(num_relation, 32) [layer.py:55] + relation_linear: Linear(32→54*32) [layer.py:53]. Each 27 relations (DIS-GEN, PHE-GEN, GEN-GEN, BPO-GEN, CCO-GEN, MFN-GEN, PWY-GEN, DIS-PHE etc, see configs/m2_trackd_relations.yaml APPROVED 2026-08-15) gets a 32-dim vector. Initially uniform random, after training learned via Adam lr 1e-3 on BCE + 64 negatives.

Worked example — Werner → WRN: Werner --[DIS-PHE]--> HP:0004322 short stature --[PHE-GEN]--> WRN. Werner_t0 = query(ASSOCIATED_WITH) = [0.8,-0.3,...32], others 0. Layer1: HP_t1 = Werner_t0 * relation(DIS-PHE) (distmult = elementwise product) aggregated via pna (mean+max+min+std then Linear). Layer2: WRN_t2 = HP_t1 * relation(PHE-GEN). After 6 layers, WRN score = MLP(WRN_t6 concat query). That's the encoding we and BioPathNet paper use identically (input_dim 32, hidden_dims [32]*6, distmult, pna, layer_norm, dependent, short_cut).

2.5 Model and training — exact config, and why not replication

- **Model:** BioPathNet b051a90 (MIT), NBFNet variant — input_dim 32, hidden_dims [32×6], message_func distmult, aggregate_func pna, num_mlp_layer 2, layer_norm, dependent true, short_cut, num_beam 10, path_topk 10, num_negative 64, criterion bce, margin 6, sample_weight false (primary), strict_negative true, filtered_ranking true, full_batch_eval true. 6-layer depth matters — >2-hop strata test the deep stack. **Patches:** scripts/patches/biopathnet-strict-negative-guard.patch (zero-candidate CUDA fix), scripts/patches/torchdrug-streaming-eval.patch

(OOM fix: streaming per-batch MRR exact, AUROC/AP batch-averaged approx).

- **Not a replication:** scripts/m1_reproduce_biopathnet.py is wiring skeleton (--dry-run only). Gate 0 two-step PrimeKG repro (torch 2.0.1/py3.10 vs 2.13/py3.12, ±10%) was **deferred to supplement** per CHANGES/0022 (owner approved 2026-08-10) to prioritize 1.43M feasibility. Mock onboarding only: reports/m2a/biopathnet_onboarding.md MRR 0.4358 CPU / 0.4194 GPU. Our results are on **SINA LODO**, not their PrimeKG random-split; not comparable.
- **Training — 1 full pass (no sub-sample):** batch_per_epoch in torchdrug/engine.py:129 means if set → sub-sample, else full. Early AD runs used batch_per_epoch 1000 (configs/m2b_taskg_run_ad.yaml) and failed (222 days/epoch if full on 9.7M). **Wave1/Step0b/Wave2** use solver.train(num_epoch=1) in scripts/pod/run_wave1_benchmark.py:406 with **no** batch_per_epoch → batch_per_epoch = len(dataloader) → 1 full pass over each unit's train2 (52-730 supervision batches) plus full 1.43M BRG message passing each batch. batch 32 effective 128/256 via accum×4/8, no NeighborLoader, Adam lr 1e-3, hidden_dims 32, BRG labels [] for KEEP.
- **Hardware:** g6e.2xlarge L40S 48GB driver 595.91 CUDA 13.2 30GB RAM AMD EPYC 7R13 (torch 2.0.1+cu118 in biopathnet py3.10, torch-toolkit 11.8 conda, DLAMI 2026.08), local 14T/50GB torch.set_num_threads(12) RSS <40GB asserted. **Gate 0 ladder:** 300→64 OOM on 48GB; 32 OK 29.28GB 3158ms 97.8 min/pass ≤140 (B1+B2 PASS), 16 15.82GB, 8 9.12GB. B3 3×2 shuffles exact, B4 88MB bitwise 0.00e+00 PASS (gate0_light.md, gate0_ladder_results.json:sha256 6c0ceace...).

2.6 Evaluation — the judges need this

- **Metrics:** filtered MRR / Hits@k (k=1,3,10,20,50,100) + MR **exact** (global ranking after micro-batching, eval_exact_check.py 9-decimal match MRR 0.007878 on Werner 730), AUROC/AP batch-averaged approx (never gated). Degree-stratified, degree-matched nulls (100 resamples).
- **Same-graph nulls:** per-unit degree (lodo_baselines_keep.md: degree 0.0006-0.0128) and PPR (0.0009-0.0170, no MRR≥0.99 on KEEP vs 1.0 on DROP for 0012853/0014527 degens) on exact training graph (train1+train2 for that unit). **Beating the best null is the result** — island band MRR 0.01-0.03 (Gate 0 island statement 29 islands 0 phenotype +1-8 → 0.01-0.03), random ~0.00006 (1/17448).
- **Statistics (pre-registered configs/m2_wave_stats.yaml):** exact Wilcoxon 2^n enumeration + hierarchical BCa n=1000 on frozen clusters + cluster-label permutation. Report All-13 and degenerate-excluded variants (frozen A3 flags all 13 primaries on KEEP — see §2.8). 1 seed = directional only; read requires seeds 1337+1339+5027.

- **Negative controls:** dual shuffles `across_disease` (gene degree preserved) + `within_disease` (disease degree preserved) on `3` probes (`0010196` Werner, `0008310` HGPS, `0007669` RCAD), `seed 1337`, `exact` marginals, `0` dups, disjoint from `valid/test` (`lodo_shuffles_keep.md`). **Must collapse `58-89%` to null band or halt** (`CHANGES/0022`).
- **Triviality `A3` (frozen `CHANGES/0021`):** per-unit `≤2-hop` distance in `train1` to test answers. Degenerate if `baseline MRR≥0.99` **or** `>10% ≤2 hops`. On `KEEP` graph `19/29` degenerate, **all `13` primaries `41.7-94.2% ≤2 hops`** (`0014157` `94.2%`, `0012853` `91%`, `triviality_audit_keep.md`). Attrition `all 13 primaries ≥10` context edges (`10-98`) vs `1-2` on `DROP`. Conflict recorded `CHANGES/0024 §(b)` — literal `A3` wipes primaries; owner decision pending; we report both variants without pre-deciding.

2.7 Pathway Discovery Funnel (`CHANGES/0025` DRAFT, `f995eef` → peer-review fix → anchored control `53f24ca`)

- **Motivation:** Gene `MRR 0.007` rank `~128` looks like miss, but `30/100` genes on one `GO:0006914 autophagy` pathway is a testable mechanism. Funnel pools `1,000` reasoning traces (`100` genes × `10` paths, `max_hops 6`) into `10` pathways.
- **Extraction:** `BioKGC/BioPathNet` gradient-weighted path interpretation (`Oleshko bioRxiv 2024 → Hu Nat Biomed Eng 2026`, we reuse verbatim) — `path_score = grad(sigmoid(score(d,g)))` `w.r.t. path edge weight × PNA aggregator weights.` `path_topk 10`, `min_path_score 0.01`, canonical dedup, `seed 1337`, `faithfulness` mask-`Top-10` control (`Jain & Wallace`).
- **Maps (already in `BRG`):** `BPO-GEN 158k` → `GO:` (∼15k), `PWY-GEN 46k` → `R-HSA-`, plus `12 Hallmarks of Aging` (`López-Otín 2013/2023`) + `MSigDB` (`configs/hallmarks_of_aging.yaml`, `docs/go_to_hallmarks.tsv` 43 rows, `≤0.15` weight, reported separately — raw `GO/Reactome` always alongside hallmarks).
- **Aggregation:** `pathway_score(P)= Σ path_score / ln(degree(P))` where `degree(P)=train1 edges touching pathway node P` `(1,431,945 KEEP)` — `Hetionet DWPC` descendant, sensitivity `log vs sqrt vs off` in supplement; `min_support 5` (`3/5/10` sens), `tie_break specificity`, `select 10`.
- **Ground truth:** held-out `test` genes only → `GO/Reactome` enrichment (`≥3` test genes per pathway); circularity disclosed (`BPO/PWY` in `train1` like `GEN-GEN` for gene task); mitigate `+ MSigDB` independent supplement. Metrics `pathway Recall@10/5, MRR, Precision@3, NDCG@10` vs `degree_null_pathway_recall` + `ORA hypergeometric Top-100` (must be beaten), `>2-hop` strata, `hierarchical BCa`.
- **Config:** `configs/m3_pathway_funnel.yaml` (`100` primary, `sensitivity [50,100,200]` pre-registered `P0`, `consensus seeds [1337,1339,5027]` mean sigmoid, `anchored` exploratory Viterbi best-path for ontology-forced paths). Guards `WRN/LMNA in Top-100` gene gate (separate), `pathway shuffle collapse` gate

before any claim, max_rss 40GB, no NeighborLoader, provenance funnel_provenance.json.
- **Prior art honesty:** TxGNN Explainer (individual), PaGE-Link (recommendation), PoPS, GSEA (flat ORA), GNNenrich, GNNExplainer/Integrated Gradients (alternatives not used — cost on 1.43M; grad×attention is citable). Novelty is aggregation architecture on LODO zero-shot OptimusKG with pre-registered nulls, not extraction.

2.8 Reproducibility — pinned

- **Pinned:** data/raw/REGISTRY.json (49 assets, SHA-256 + size), data/raw/manifests/*.json (m2_build_lodo_exports, m2_lodo_exports_keep, m2_lodo_shuffles_keep, m2_lodo_baselines_keep, wave1_benchmark_results*.manifest.json, shuffle_control_results.manifest.json, gate0_ladder_results.manifest.json, m3_pathway_funnel.json), configs/m2_trackd_relations.yaml:status APPROVED.
- **Code:** github.com/Abolfazl-Alipour/sina-platform main (this paper), BioPathNet b051a90 MIT (third_party/biopathnet + third_party/torchdrug 6066fbd), sina py3.12 + biopathnet py3.10 (environment.lock.yml, environment.biopathnet.cu118.yml). Patches auto-applied via scripts/cloud_bootstrap.sh. Test + lint counts are re-measured at each revision (reports/REPRODUCTION_2026.md is the live count of record).
- **EBS persist:** scripts/pod/persist_snapshot.sh snapshots /opt/sina/data/processed, REGISTRY.json, all checkpoints_* (step0b/wave1/shuffle 31 .pth 88-105MB + Wave2 16 ≈88MB), reports/, ~/.kimi-code/.gemini → /home/ubuntu/persist 17GB; persist_restore.sh restores 1-2 min on fresh g6e.2xlarge. Checkpoints gitignored — EBS is source of truth. Compressed sina_full_backup_v2_2026-08-27.tar.gz 5.5GB is off-host backup.

2.9 Completion track (M2c) — exact reproduction (CHANGES/0027, g7e.2xlarge Blackwell sm_120, 2026-09-05→06)

- **Why a second track:** the LODO benchmark (§3.3–3.4) measures zero-shot transfer; M2c measures the complementary exam — recover held-out associations for a seen disease (completion: hide 5% of DIS-GEN, train on the rest). Same frozen model (b051a90 32×6 distmult/PNA, Adam lr 1e-3, pinned 17,448 universe, seed 1337), different supervision geometry. Two diseases: PD mondo__0005180 (train2 190,647 / valid 967 / test 339) and AD mondo__0004975 (189,450 / 1,882 / 621).
- **Exports:** data/processed/m2c/completion_exports/<unit>/ (train1.txt 1,431,945 KEEP-only — the ANA-GEN 1.4M expression edges were removed after three inertness proofs; splits byte-identical to the ANA build). Pinned list data/processed/m2c/

pinned_genes_17448.txt (both units agree; all split tails ⊆ pinned).

- **Loss (the 0027 switch):** per-row full filtered softmax over all 17,448 scores (1 held-out positive + every allowed negative, knowns masked, log-softmax temp 1.0, NLL) replaces multi-hot BCE; Probe 2 (reports/m2c/objective_probe.md, 200 identical batches) showed +35% train-window MRR and -11% step time. Aux 1:1 GEN→BPO/PWY and filtered val unchanged. **Known deviation (numbers unaffected):** on KEEP exports the relation/type fallback misroutes aux to GO cellular_component 94,150 + HPO phenotype→gene 768,844 instead of GO-BP + reactome (m2c_arm_e_train.py:315-316); aux never enters val/test. Default resolution is document-only unless the owner orders otherwise.
- **Epochs:** PD 1,907 batches 8.84h, AD 1,895 batches 9.02h, B=100, ≈90.5GB, single epoch after flat-line evidence (val MRR flat ~0.015 PD / ~0.011 AD all epoch). Checkpoints epoch_001/best/last.pth are **tracked in git** (reports/m2c/checkpoints_completion_noana/<unit>/, ~5MB each) — unlike M2b's gitignored EBS-only checkpoints.
- **Test-once protocol:** hidden test scored exactly once per checkpoint (scripts/pod/m2c_pd_test_once.py, filtered over pinned 17,448, optimistic-tie rank, bootstrap 1000/1337), bpn120 env with CUDA_VISIBLE_DEVICES="" (zero GPU memory; the sina env lacks ogb — runbook shorthand corrected in HANDOFF.md 2026-09-06 entry). Genome-wide cross-check in the OT rescore reproduces test-once bit-for-bit.
- **External panels:** Open Targets association panels re-pulled in-repo (m2c_ot_panel_eval.py pull, reports/m2c/ot_panels/, coverage byte-identical on re-pull); two-stage = P(in panel) × recall@10 given coverage; circularity audit at evidence-class level (circularity_audit.json — per-edge provenance is unannotated in OptimusKG, stated assumption). Candidate generators attacked separately on CPU (m2c_generator_wave2.py, generator_panels_wave2.md).
- **Temporal + functional probes:** GWAS Catalog bulk via EBI FTP (DisGeNET bulk is login-walled) → year-stamped loci (m2c_gwas_loci_build.py, cutoff 2020); prospective null (m2c_prospective_null.py) and prospective/per-locus model (m2c_prospective_model.py on pinned-order score dumps, order asserted); VEP strict-coding singletons as functional gold (m2c_vep_culprits.py, Ensembl REST, pre-registered power rule: pooled novel n≥15 or no test run).

Repro commands (copy-paste):

```
git checkout 53f24ca
conda run -n sina python -m sina.lint_reports reports/ --config
        configs/report_lint.yaml
conda run -n sina python -m pytest tests/unit -q
python scripts/m2_build_lodo_exports.py --config configs/
        m2_lodo_exports_keep.yaml --resume
```

```
conda run -n biopathnet python scripts/pod/gate0_ladder.py --
        config configs/m2b_gate0_ladder.yaml
conda run -n biopathnet python scripts/pod/run_step0b_ablation.py
        --batch-size 32 --seed 1337
conda run -n biopathnet python scripts/pod/run_wave1_benchmark.py
        --batch-size 32 --seed 1337  # + 1339/5027
conda run -n sina python scripts/m2_wave_stats.py --config
        configs/m2_wave_stats.yaml
python scripts/pod/extract_paths.py --config configs/
        m3_pathway_funnel.yaml --disease mondo:0010196
python scripts/pod/anchored_topology_null.py --config configs/
        m3_pathway_funnel.yaml
```

2.10 External head-to-head vs Open Targets L2G — train-knowns vs test-knowns (2026-09-11→25)

- **Setup:** same GWAS locus definitions as the per-locus probe (§3.7), `scripts/pod/m2c_loci_headtohead.py`, pessimistic-tie ranking, locus-blocked bootstrap `1000/1337`. Per locus with ≥2 genes and ≥1 known gene: top-1 / top-5 of {model, OT L2G, degree} at recovering a known gene. Verdict rule: paired diff beyond ±0.05 with CI excluding 0 = WIN/LOSS, else TIE.
- **The split that matters:** default grades on known genes present in the model's `train2` supervision (**train-knowns** — memorization-compatible); `--known-split test` grades only on known genes absent from `train2` (**test-knowns** — the leakage-controlled exam). A uniform-gene-pick chance-draw control (`100` draws) is reported alongside so "beats OT" can be checked against "beats chance."
- **Artifacts:** full panel `reports/m2c/loci_headtohead_mondo__{0005180,0004975}.json` (+ `.md` summary, + manifests); test-knowns `reports/m2c/loci_headtohead_mondo__{0005180,0004975}_testknown.json` (+ `.md`, + manifests).

2.11 Tabular and functional-evidence controls (M4B / M5B, CPU, 2026-09-12→26)

- **M4B (tabular, `scripts/m4b/tabular_baseline.py`):** gradient-boosted trees (LightGBM/XGBoost) on per-gene locus features over OT gold-standard + FLAMES loci. Views: `CV` (5-fold), `TMP` (temporal cutoff), and the **covered subset** (`n=491` OT-scored loci — the fair fight; the full-set OT number is a coverage artifact because OT auto-misses `1,326` UK Biobank loci). Arc v1→v3 documented in `reports/m4b/tabular_baseline{,_v1,_v2,_v3}.md`; **v4 = fresh-features rebuild** (no current-graph features, no OT scores as inputs) with chromosome- and gene-blocked folds, evaluated as the M5B base arm.

- **FLAMES replication (inside M4B v3):** their shipped model + their loci, our top-1 metric — a direct test of the published "FLAMES beats OT" precedent.
- **M5B (functional tracks, `m5b/`):** pre-registered arms add fine-mapping posteriors, ClinVar, regulatory elements (cCRE), and eQTL overlaps to the v4 base. Deciding metric: covered PR AUC diff vs OT L2G, locus-blocked bootstrap. HPO arm governed by the pre-registered inner/outer rule (outer trails inner by `>5pts` → keep the simpler model). Verdict `m5b/reports/verdict.md`; raw `m5b/reports/v5_results.json` (+ `hpo_results.json`, track manifests).

---

3. Results

3.1 Gate 0 Light (L40S, `gate0_light.md 2026-08-15, gate0_ladder_results.json:sha256 6c0ceace...`)

`Werner 0010196` (`59k train2 + 1.42M BRG`): `300→64 OOM` on `48GB`; **`32 OK 29.28GB 3158ms 97.8 min/pass ≤140`** (`B1+B2 PASS`), `16 15.82GB 106.4`, `8 9.12GB`. `B3 3×2` exact, `B4 88MB` bitwise `0.00e+00`, `B6` island statement `29 islands 0 phenotype +1-8 → 0.01-0.03`.

Full ladder:

Batch	Status	VRAM GB	ms/batch	min/pass
300	OOM	45.51	-	-
280	OOM	42.52	-	-
250	OOM	45.95	-	-
200	OOM	43.77	-	-
150	OOM	42.69	-	-
100	OOM	41.99	-	-
64	OOM	44.45	-	-
32	**OK**	**29.28**	**3158.1**	**97.8**
16	OK	15.82	1718.6	106.4
8	OK	9.12	1001.6	124.1

3.2 Step 0b (3 probes, 12 arms, `bfloat16 1337, 5005e7d, reports/m2b/step0b_ablation.md`)

probe	`drop_p1`	`drop_p2`	`keep_p1`	`drop_capped`
Werner `0010196`	0.00346	0.00347	**0.00653**	0.00444
HGPS `0008310`	0.00555	0.01230	**0.01015**	0.00621

probe	drop_p1	drop_p2	keep_p1	drop_capped
RCAD 0007669	0.00629	0.00425	**0.00930**	0.01056

KEEP +73.2% mean (88.7/82.9/47.9%), 1 pass frozen (pass2 flat/degrade), unweighted primary.

3.3 Wave 1 — 13 primaries (KEEP 1,431,945, 1 pass, batch 32, f72b9a2, 24.1h, reports/m2b/wave1_benchmark.md, wave_stats.md)

One-seed 1337: mean MRR 0.00781 (130× random ~0.00006, lower edge 0.01-0.03), 6/13 vs best degree/PPR null, p=0.376 Wilcoxon directional.

unit	test	model MRR	degree	PPR	lift	≤2-hop%
0100164 PNDM	459	**0.01662**	0.0007	0.0144	1.16×	47.9%
0008199 PD1	57	0.01425	0.0016	0.0170	0.84×	73.7%
0011913 MADA	68	0.01292	0.0057	0.0157	0.82×	64.7%
0008310 HGPS	394	0.00918	0.0070	0.0022	1.31×	64.0%
0007669 RCAD	258	0.00811	0.0015	0.0137	0.59×	90.7%
0010196 Werner	730	0.00788	0.0062	0.0031	1.26×	83.8%
0012853 MAD-B	201	0.00681	0.0128	0.0017	0.53×	91.0%
0012192	281	0.00582	0.0075	0.0009	0.78×	78.6%
0010785	96	0.00511	0.0014	0.0082	0.62×	77.1%
0014157	52	0.00470	0.0018	0.0018	2.54×	94.2%
0012522	251	0.00455	0.0013	0.0019	2.40×	45.0%
0100165	329	0.00313	0.0007	0.0028	1.13×	49.2%
0957576	127	0.00243	0.0006	0.0031	0.78×	41.7%

Three-seed 1337+1339+5027 (53f24ca, WIP 10 seed5027 13/13 checkpoints merged): mean 0.00775 ±0.00008 [0.00754,0.00796], Wilcoxon 31.0 p=0.3396, BCa 0.00775 [0.00548,0.01101], cluster p=0.938, <2 non-degenerate (A3 literal), per-unit 0008199 0.01639 / 0100164 0.01624 top. Confirms 1337 — still not significant.

3.4 Wave 2 — 16 secondaries (seed 1337, KEEP, 2026-08-26T23:08:35Z, phase_d_wave2_secondary_benchmark, wave2_secondary_results.json)

16 units: 5 secondary-evidence + 11 secondary-low-context, 0=5%.
Sorted by test MRR:

unit	name	test MRR	H@10	H@100	valid MRR	vs best null
mondo:0018911	maturity-onset diabetes of the young	**0.04142**	0.042	0.182	0.01587	loses to PPR 0.0799
mondo:0011150	acroosteolysis-keloid-like	0.02230	0.062	0.208	0.00801	**BEAT** PPR 0.0068
mondo:0013523	Nestor-Guillermo progeria	0.01780	0.031	0.125	0.01341	loses to PPR 0.0358
mondo:0018203	LMNA cardiocutaneous	0.01015	0.015	0.103	0.00575	**BEAT** degree 0.0062
mondo:0017279	young-onset Parkinson	0.00885	0.005	0.064	0.01652	loses to PPR 0.0092
orphanet:75496	Ehlers-Danlos progeroid	0.00651	0.000	0.049	0.00856	loses to PPR 0.0079
mondo:0000828	juvenile Parkinson	0.00632	0.006	0.041	0.00210	loses to PPR 0.0087
mondo:0030089	diabetes permanent neonatal	0.00524	0.010	0.041	0.00221	**BEAT** PPR 0.0024
mondo:0015140	early-onset AD	0.00487	0.010	0.035	0.00581	loses to PPR 0.0057
efo:0005422	skin aging	0.00466	0.005	0.028	0.00342	loses to degree 0.0157
mondo:0100087	familial Alzheimer	0.00417	0.009	0.030	0.00493	loses to degree 0.0052
mondo:0007088	Alzheimer type 1	0.00366	0.000	0.037	0.00838	

unit	name	test MRR	H@10	H@100	valid MRR	vs best null
						loses to PPR 0.0090
mondo:0019321	atypical Werner	0.00352	0.000	0.040	0.00260	**BEAT** PPR 0.0020
mondo:0014527	progeroid-hepatocellular	0.00296	0.000	0.038	0.00202	loses to degree 0.0100
mondo:0012480	diabetes transient neonatal 2	0.00294	0.003	0.017	0.00202	loses to PPR 0.0189
mondo:0020525	transient neonatal diabetes	0.00132	0.001	0.010	0.00222	loses to degree 0.0035

Mean 0.00917 median 0.00506 (lodo_baselines_keep.md degree/PPR per unit). 4/16 beat best null (generous), Wilcoxon p=0.9673 — no claim (even best 0018911 0.04142 loses to PPR 0.0799). All KEEP 1,431,945, seed 1337 only. Full per-unit valid/test in reports/m2b/wave2_secondary_results.json + wave2_secondary_results.manifest.json (SHA-256).

3.5 Shuffle collapse (3 probes ×2, 267f85e, shuffle_control_benchmark.md)

probe	mode	MRR	H@10	true	collapse
Werner	across	0.00157	0.00%	0.00788	**-80.1%**
Werner	within	0.00238	0.27%	0.00788	**-69.8%**
HGPS	across	0.00149	0.00%	0.00918	**-83.8%**
HGPS	within	0.00192	0.13%	0.00918	**-79.1%**
RCAD	across	0.00340	0.19%	0.00811	**-58.1%**
RCAD	within	0.00090	0.00%	0.00811	**-88.9%**

58-89% collapse, Hits@10→0.0-0.27% — zero leakage.

3.6 Pathway Discovery Funnel (`f995eef` spec → 2127-line `extract_paths.py` at `53f24ca` → anchored control `53f24ca`)

Frozen funnel (*Top-100 grad×PNA → 10* pathways *log(degree) min_support 5*, *K=[50,100,200]* sens, *>2-hop* strata, *faithfulness* mask, *WRN/LMNA* gate, *ORA* baseline): `reports/m2b/pathway_funnel.md` for `0008310` (and `0010196`) — **frozen `Recall@10 0.0` vs `degree-null 0.0069` vs `ORA 0.0027` (true `1459, >2-hop 0.0`)**, `K=50/100/200` all `0.0`, `faithfulness VACUOUS — no paths to mask`. **Unconstrained beam bypasses `GO/PWY` ontology nodes (`EXPLORATORY` Viterbi best-path table shows `GO:0010628` hits but is not the frozen endpoint; beam `~40h/disease` infeasible).** Anchored control `53f24ca` (`anchored_topology_null.md`):** exploratory anchored table (`support/ln(degree)` null `0.0043 vs 0.0043` Werner `Jaccard 0.82 9/2085` hits, `0.0069 vs 0.0062` HGPS `0.54 10 vs 9/1459` hits) **ties the topology null** and survives `within` shuffle (`0.0069→0.0062`) while gene `MRR` collapses `58-89%` — reveals artifactual `GO` ranking, not learned signal; `across` 100% collapse `PASS` but `within ~10% FAIL`. No recovery claim. Figures `pathway_figure*.png` (`20-58K`) land but metric is `0.0` — do not quote funnel in grant until `Recall@10 > degree-null` and shuffle collapses. **This is the honest state at `53f24ca`; compute queue is done (`FRESH_INSTANCE_HANDOFF.md`).**

Anchored detail: candidates `2578` (Werner) / `2220` (HGPS) with direct `train1` annotation edge to consensus `Top-100` genes, prefiltered `support ≥3` → `384/301`; truth `2085/1459` pathways from held-out test genes (`≥3` genes each). Null = `support/ln(degree)` over identical candidate set. Per-candidate weights not persisted (`data/processed/m3/` only Top-10 tables survive) — comparison is metric-level + Top-10 Jaccard.

3.7 Completion track — disease-gene recovery on OptimusKG (`CHANGES/0027, 2026-09-05→06, reports/m2c/completion_wave.md`)

No-ANA rank-loss full epochs (*KEEP* graph *1,431,945* train1, *rank* full filtered softmax, pinned *17,448* universe, *seed 1337*, *B=100*, single epoch after flat-line evidence): PD (`mondo__0005180`, `339` test, `8.84h`, `90.46GB`-class) **test `MRR 0.01887 [0.01297,0.02704] H@10 0.02950`** vs `degree` floor `0.00818` (`2.3×`) vs supervision-only null `0.01484` (`+27%`, `H@10` identical `10/339`); AD (`mondo__0004975`, `621` test, `9.02h`, `1895` batches, `90.46GB`, val `MRR` flat `~0.011` all epoch) **test `MRR 0.01061 [0.00735,0.01506] H@10 0.01610`** vs floor `0.00909` (`+17%` point, `CI` overlaps) vs null `0.00975` (`+9%` point, `H@10` identical `10/621`). Stack-invariant: `±ANA`, `BCE/rank`, width `32/64`, batch `32/100`, disease — every road lands on its disease's plateau. **Two-stage rerank** (`OT` panels, `K=100/200/500`): PD `0.0295` = `OT` null at all `K`; AD `0.0161` vs `OT` null `0.0145` — hit set panel-independent across `10+` panels, conversion not fixable; generation fixable (`union_best` coverage

49.3%@500). **Prospective** (GWAS Catalog bulk, cutoff 2020): null PRE collapses (0/82 PD, 0/1086 AD); model on clean future genes PD MRR 0.00373 (0/29) / AD MRR 0.00300 (2/239) — discovery certified dead twice. **Per-locus recovery** (multi-gene loci w/ known): PD model top-1-known 0.904 vs degree 0.615 vs chance 0.677; AD 0.954 vs 0.816 vs 0.734 — first task beating popularity wide, but knowns are train positives (recovery, not discovery; §3.8 grades the same panel against Open Targets L2G and re-grades it on unseen knowns). **Functional final test** (VEP strict-coding singletons): pooled novel n=2 vs 15 required — power fails, no test run, discovery UNDECIDED; anecdotes PD 1/5 top-1 (a novel culprit), AD 6/9 (all known; APOE control 2nd/49, degree 1st). Claim: disease-associated gene recovery with a certified-negative discovery boundary — the ceiling is the supervision, on both diseases.

3.8 Head-to-head vs Open Targets L2G — and the test-knowns correction (reports/m2c/loci_headtohead.md, 2026-09-11→25)

Train-knowns panel (known genes present in *train2*; memorization-compatible):

disease	n	model top-1	OT L2G	degree	chance-draw	model−OT diff	v
PD 0005180	52	**0.9038** [0.8269,0.9808]	0.6154	0.6154	0.677	**+0.2885** [0.1731,0.4231]	**W**
AD 0004975	239	**0.9540** [0.9247,0.9791]	0.5356	0.8117	0.739	**+0.4184** [0.3515,0.4896]	**W**

Test-knowns panel (*--known-split test*: only known genes never in *train2*; the leakage-controlled exam):

disease	n	model top-1	OT L2G	degree	chance-draw	model−OT diff	v
PD 0005180	10	0.300 [0.100,0.600]	**0.600**	0.500	0.330	**−0.300** [−0.600,−0.100]	**L**
AD 0004975	31	0.1613 [0.0323,0.2903]	0.2258	0.2903	0.328	−0.0645 [−0.2581,0.1290]	**-**

Read together: the model's per-locus advantage exists only over genes it was trained on. On unseen knowns it loses to (PD) or ties (AD) Open Targets L2G, and both test-knowns scores sit **below their own chance-draw means** — the model's ranking carries no usable signal for genes outside its supervision. PD's test-knowns n=10 is small and stated plainly; AD's n=31 tie has a CI wide enough to include a real loss. The 0.904/0.954 numbers remain true as train-positive recovery and are quoted nowhere in this paper without that qualifier.

3.9 Tabular control (M4B) and the FLAMES replication (`reports/m4b/, 2026-09-12→26`)

Covered subset (*n=491* OT-scored loci — the fair fight): GBM top-1 `0.5927 [0.5540,0.6314]` vs OT `0.5866 [0.5397,0.6273]` → paired diff `+0.006 [−0.039,+0.053]` **STATISTICAL TIE** (structural note: OT trained on all these gold loci — home advantage; our GBM sees `4/5` folds). vs nearest-gene distance `0.5356` → `+0.057 [+0.014,+0.104]` — **first significant win over the distance floor** on this panel. Full-set GBM `0.4491` vs OT `0.1723` is a coverage artifact (OT auto-misses `1,326` UK Biobank loci) and is not a claim. Temporal view cannot fire: zero post-2020 loci exist in the positives (gold curation ends 2020) — the pre-registered comparison is `N/A`, itself a finding. FLAMES replication (their shipped model, their loci, our top-1): raw XGB `0.602` vs OT `0.689` → `−0.087 [−0.165,−0.010]` (**OT wins clearly**, `n=103`); full FLAMES `+0.058 [−0.019,+0.136]` (`n=103`) and `+0.064 [0.000,+0.129]` (`n=140`) — leads by `+2` to `+6pts`, **never significant at 95% on top-1**. The published "FLAMES beats OT" precedent does not survive its own shipped benchmark tables on this metric — a citable correction. v4 fresh-feature rebuild (no current-graph features, no OT-score inputs; chr/gene folds): evaluated as the M5B base arm — PR AUC `0.253` vs OT `0.4767` (§3.10) — **LOSS**.

3.10 Functional-evidence arms (M5B, `m5b/reports/verdict.md, 2026-09-26`)

Deciding metric: covered PR AUC diff vs OT L2G (`0.4767`, `n=491`), locus-blocked bootstrap CI. Arms (chromosome folds; gene folds worse everywhere):

Arm	PR AUC	Diff vs OT + 95% CI	Top-1 diff
v4 base (fresh-feature, no graph)	0.253	−0.222 [−0.271,−0.176]	−0.134
+ fine-mapping overlaps	0.253	−0.222 (no gain)	−0.165
+ ClinVar	0.276	−0.198 (+0.024, only pulse)	−0.143
+ regulatory elements	0.250	−0.225 (no gain)	−0.153
+ eQTL overlaps	0.253	−0.221 (no gain)	−0.157
all five tracks	0.300	−0.174 [−0.225,−0.126]	−0.128

Arm	PR AUC	Diff vs OT + 95% CI	Top-1 diff
HPO/model-selected	0.255	discarded per pre-registered rule (outer 0.255 trails inner 0.54-0.60 by >5pts → fixed LightGBM kept)	—

Every CI excludes zero on the negative side — **LOSS on all arms**. What each track earned: fine-mapping ~0.000 (distance already says what posteriors say); ClinVar +0.024 (only measurable pulse); regulatory ~0.000; eQTL ~0.000 at 1.4% row coverage (exact-match sparsity); combined +0.047 over base against a 0.22 gap. Bottom line, verbatim from the verdict: five public datasets, honest splits, tuned and untuned — our best is 0.300 against OT's 0.477. OT's functional-evidence advantage survives everything legally addable on CPU; beating it requires coverage OT lacks, future (prospective) discoveries, or a modeling breakthrough — not more public features.

---

4. Discussion — what survives hostile review

- **Survives:** 1.43M feasibility, KEEP +73.2% ablation, 58-89% shuffle collapse, fidelity 9-decimal, pre-registered Wilcoxon/BCa/cluster framework, positive-control gate, cost-validated 24.1h Wave 1, first OptimusKG gene benchmark, honest island band 0.01-0.03 and A3 disclosure, encodings identical to BioPathNet paper (see §2.4) so comparison is fair on protocol not implementation.
- **Survives (completion track, §3.7):** full-epoch disease recovery beats the same-graph floor on both diseases (PD 2.3×, AD +17% point) with H@10 pinned to the counting null; stack-invariance (±ANA, BCE/rank, width, batch, disease); prospective collapse is replicated for null and model (second independent no-discovery certificate); the functional final test was not run when power failed (2 vs 15) — the rule held against the temptation to sell anecdotes.
- **Survives (external head-to-head, §3.8):** on train-knowns the model beats OT L2G and popularity wide (PD +0.29, AD +0.42) — memorization-grade recovery, quoted only with that qualifier. On test-knowns the margin inverts (PD LOSS 0.300 vs 0.600; AD TIE 0.1613 vs 0.2258, both below chance-draw means). The correction is itself a deliverable: the harness caught its own headline.

- **Survives (tabular/functional controls, §3.9-3.10):** a leakage-controlled LightGBM ties OT L2G on covered loci (`+0.006`) and beats nearest-gene distance significantly (`+0.057`) — CPU features match the graph model's external comparator with no graph at all; the shipped FLAMES benchmark's OT win does not replicate on top-1 (literature correction); five functional-evidence tracks all lose to OT with negative CIs throughout.
- **Does not survive as a claim:** `p≈0.34` with `3` seeds is not significant (`6/13` is coin toss), `A3` wipes primaries (`13/13` degenerate on `KEEP`), funnel `0.0` is not `> degree-null` (and anchored ties null `0.0043/0.0069` `Jaccard 0.82/0.54`), `mask` faithfulness is vacuous, `Top-100` now pre-registered `50/100/200` — ground truth circularity (`BPO/PWY` in `train1`) now tested by `anchored` vs `topology null` (`support/ln(degree)`) and shown as artifact (`within` survives); `across` shuffle gate `PASS` but `within ~10% FAIL`.
- **Does not survive as a claim (completion + external):** genome-wide discovery on this supervision (prospective `MRR ≈ 0.003` both diseases, `H@10 ≈ 0`); rerank conversion beyond the counting null (`H@10` identical both diseases); per-locus causal-gene claims (recovery ≠ causality — twice the top-ranked known gene was not the functional culprit, and the final test never reached power); the per-locus "rank the knowns" product hypothesis (test-knowns re-grade: no unseen-gene signal, §3.8); any functional-track tabular claim against OT (M5B, §3.10).
- **Encoding note:** Both SINA and BioPathNet use **query-indicator node init + learned relation embeddings (32-dim, distmult) + 6×PNA layers** (§2.4). The failure is not encoding — it's task difficulty: `LODO zero-shot` (`hide whole disease`) vs `random edge split` (`hide 10% edges, 90% of disease seen`). On random split BioPathNet gets `MRR 0.547` (`gene function`) / `0.36` (`synthetic lethality`); on LODO we get `0.007`. Different exams.
- **The pivot is right but not novel in parts:** path extraction is `BioKGC/BioPathNet` reuse, hub penalty is `Hetionet DWPC`, enrichment is `GSEA/ORA` — novelty is aggregation architecture on LODO zero-shot `OptimusKG` with pre-registered nulls and why it matters (pools weak `rank 50-150` genes on one `GO:0006914` pathway → lab-testable `3×3` hypothesis, not phone book `1000`).

Fundable framing: Leakage-certified hypothesis-triage funnel, not single-gene discovery — *`KEEP`* validated, *`1 pass`* validated, *`6-layer`* topology proven by shuffle collapse, pathway funnel gating on shuffle collapse before any claim — with the external head-to-head correction (§3.8) as the proof that the harness falsifies its own headlines.

---

5. Limitations

1. A3 literal excludes all 13 primaries on KEEP; owner decision CHANGES/0024 §(b) pending — report both All-13 and degenerate-excluded.
2. 3-seed p≈0.34 — directional only; Wave 2 16 secondaries (mean 0.00917, 4/16 beats null, p=0.9673) is complete (2026-08-26T23:08:35Z, 53f24ca) — still no claim; seeds 1339/5027 primary CI WIP (10 seed5027 13/13 checkpoints merged) needs final tag for 3-seed CI tightening.
3. Funnel 0.0 + anchored ties null (0.0043/0.0069) — unconstrained beam bypasses ontology nodes and exploratory anchored is topology artifact (within survives 0.0069→0.0062); needs constrained beam through BPO/PWY nodes (not ~40h exhaustive) or hallmark-only ground truth before quoting.
4. AUROC/AP batch-averaged approx; absolute MRR 0.007-0.008 lower edge of 0.01-0.03; Top-100 grad vs uniform and log vs sqrt sensitivity still supplement.
5. Not a replication of BioPathNet paper — PrimeKG 8.1M random-split MRR 0.3-0.5 not rerun; SINA is a harder LODO exam on 1.43M OptimusKG.
6. Cut graph + 1 pass only — ANA-GEN 8.79M dropped, full 21.8M not trained; NEIGHBORLOADER disabled per design; multi-GPU full graph left for future.
7. Completion track is one seed, one epoch per disease — plateau claims rest on invariance across knobs, not seeds; AD's margin over floor/null is point-only (CIs overlap). The no-ANA aux misroute (§2.9) is documented, not fixed — a fixed-aux rerun could move val dynamics (test is aux-free, so published numbers stand regardless).
8. Discovery is certified dead only for this supervision + genome-wide ranking; per-locus causal-gene claims are UNDECIDED (power failure, not a negative).
9. The per-locus recovery headline (0.904/0.954) is train-positive memorization: the test-knowns split shows no edge over OT and sits below chance-draw means. PD's test-knowns n=10 is a tiny sample (stated plainly); AD's n=31 TIE has a CI that includes a real loss.
10. M5B is single-run, one seed family (1337/1339/5027); eQTL overlap is exact-variant-match only (1.4% row coverage, no LD expansion, no ABF); the prospective 4,104-locus set is curated but not yet evaluated.
11. Every external comparison rests on one seed / one epoch per disease on the graph side; OT's gold-set home advantage in M4B is disclosed but not fully correctable; OT coverage artifacts bound the full-panel head-to-head.

---

6. Data and Code Availability — complete file inventory for a judge

A judge needs these — every file is hash-pinned or `git` tracked, nothing hidden:

Code (git `main`, `MIT` — M2b sections pinned at `53f24ca`, M2c at `ab23259`): * Repo `github.com/Abolfazl-Alipour/sina-platform` `main` (this paper), `BioPathNet` `b051a90` `MIT` at `third_party/biopathnet` + `torchdrug` `6066fbd` at `third_party/torchdrug`, `sina` `py3.12` CPU + `bpn120` `py3.10` `torch 2.7.1+cu128` (Blackwell `sm_120`; frozen `cu118` env cannot run there — `CHANGES/0026`). GPU env line in `HANDOFF.md` §1; CPU evals run `bpn120` with `CUDA_VISIBLE_DEVICES=""`. * Patches auto-applied: `scripts/patches/biopathnet-strict-negative-guard.patch`, `scripts/patches/torchdrug-streaming-eval.patch` (via `scripts/cloud_bootstrap.sh`). * Core scripts: `scripts/m2_build_trimmed_taskg.py` (BRG fix `dst_id`), `scripts/m2_build_lodo_exports.py`, `scripts/m2_build_crosswalk.py`, `scripts/m2_build_splits.py`, `scripts/m2_lodo_audits.py`, `scripts/m2_lodo_shuffles.py`, `scripts/m2_run_lodo_baselines.py`, `scripts/m2_triviality_audit.py`, `scripts/m2_wave_stats.py`, `scripts/m2_wave1_report.py`, `scripts/pod/run_wave1_benchmark.py`, `scripts/pod/run_step0b_ablation.py`, `scripts/pod/gate0_ladder.py`, `scripts/pod/gate0_resume_drill.py`, `scripts/pod/extract_paths.py` (2127 lines), `scripts/pod/anchored_topology_null.py`, `scripts/pod/eval_exact_check.py`, `scripts/pod/persist_snapshot.sh`, `scripts/pod/persist_restore.sh`. * Tests: `tests/unit/*.py` — count re-measured at each revision (`reports/REPRODUCTION_2026.md`); includes `test_m2c_loci_headtohead.py` (head-to-head + test-knowns guards). * Linter: `sina.lint_reports` with `configs/report_lint.yaml` (forbidden terms per §13.11 — see config) — `0` violations (count of record in `reports/REPRODUCTION_2026.md`). * Completion scripts (`scripts/pod/`, all `ruff`-clean): `m2c_arm_e_train.py` (rank-loss wave runner, `--exports-dir/--ckpt-root/--probe-loss/--probe-k`), `m2c_pd_test_once.py` (hidden-test-once, `--unit/--exports-dir/--ckpt-root`), `m2c_ot_panel_eval.py` (`pull/score/union/circularity`, `--dump-scores-dir`), `m2c_generator_wave2.py` (7 candidate generators), `m2c_gwas_loci_build.py` (GWAS bulk → year-stamped loci), `m2c_prospective_null.py` / `m2c_prospective_model.py` (Probes 0/A/B), `m2c_reweighted_null.py` (6-variant label-reweight gate), `m2c_vep_culprits.py` (VEP strict-coding singletons), `m2c_rerank_probe.py` / `m2c_shortlist_generator_probe.py` / `m2c_prune_junk_loci.py` (0 junk found — the `43k` were ANA passengers), `m2c_loci_headtohead.py` (OT head-to-head + `--known-split test`). * Tabular/functional scripts: `scripts/m4b/tabular_baseline.py` (M4B views + FLAMES replication), `m5b/` pipeline + manifests (`m5b/reports/*.manifest.json`).

Configs (hash-pinned, `SHA-256` in manifests): `configs/m2_trackd_relations.yaml` `APPROVED 2026-08-15`, `configs/m2_lodo_exports_keep.yaml`, `configs/m2_lodo_shuffles_keep.yaml`,

configs/m2_lodo_baselines_keep.yaml, configs/m2_wave_stats.yaml, configs/m3_pathway_funnel.yaml (with top_k 100 [50,100,200], path_topk 10, max_hops 6, grad×PNA, min_support 5, log(degree)), configs/m2b_gate0_ladder.yaml, configs/disease_families.yaml, configs/hallmarks_of_aging.yaml, configs/m1_biopathnet_repro.yaml (deferred).

Data (EBS 17GB + Registry): * data/raw/REGISTRY.json (49 assets, SHA-256 + size per asset: OptimusKG 37 files, PrimeKG 8.1M, MONDO 53MB, TxGNN splits 68 diseases, AnyBURL 23-1x). * data/raw/manifests/*.json (19 manifests: m2_build_lodo_exports, m2_lodo_exports_keep, m2_lodo_shuffles_keep, m2_lodo_baselines_keep, wave1_benchmark_results, shuffle_control_results, gate0_ladder_results:sha256 6c0ceace..., m3_pathway_funnel, etc). * data/processed/m2/lodo_exports/<unit>/ (29 units, each train1.txt 1,431,945, train2.txt, valid/test, entity_*.txt, strip_list.json, manifest.parquet). * data/processed/m1/gene_properties.parquet (61,306 genes for annotation).

Reports (judge reads first, in order): HANDOFF.md (live journal, read first), reports/SINA_PAPER.md (this file, single record), reports/REPRODUCTION_2026.md (clean-machine verification log), reports/m2c/loci_headtohead.md + loci_headtohead_mondo__*_testknown.md (the external verdict and its correction), reports/m2c/completion_wave.md (completion verdict + AD/PD numbers), reports/m2c/noana_rank_epoch.md (PD epoch), reports/m2c/prospective_per_locus.md (Probes 0/A/B/A2/B2), reports/m2c/gold_locus_test.md (VEP final test + anecdotes), reports/m2c/reweighted_null.md (label-reweight gate), reports/m2c/generator_panels_wave2.md (generation attack), reports/m2c/objective_probe.md (Probe 2 rank-vs-BCE), reports/m2c/rerank_probe_pd.md + shortlist_generator_probe.md (generation bottleneck), reports/m2c/peer_review_status_2026-09-03.md (superseded by §3.7-3.10/§10 — kept as audit trail), reports/m4b/tabular_baseline_v3.md (covered-subset tie + FLAMES replication), m5b/reports/verdict.md (LOSS all arms), then the M2b chain: reports/m2b/gate0_light.md (ladder 29.28GB), reports/m2b/lodo_exports_keep.md (29 units), reports/m2b/lodo_baselines_keep.md (degree 0.0006-0.0128 PPR 0.0009-0.0170), reports/m2b/lodo_shuffles_keep.md (3×2 shuffles), reports/m2b/wave_stats.md (mean 0.00775 p=0.3396 BCa [0.00548,0.01101] cluster p=0.938), reports/m2b/wave1_benchmark.md, reports/m2b/wave2_secondary_results.json (16 units, mean 0.00917 p=0.9673), reports/m2b/shuffle_control_benchmark.md (58-89% collapse), reports/m2b/pathway_funnel.md (Recall@10 0.0 vs 0.0069), reports/m2b/anchored_topology_null.md (0.0043/0.0069 ties), reports/m2b/triviality_audit_keep.md (13/13 degenerate 41.7-94.2%), reports/m2b/gate0_ladder_results.json (6c0ceace), CHANGES/ (0020 freeze 29=13+5+11, 0025 funnel, 0024 A3 conflict, 0026 Blackwell divergence,

0027 no-ANA rank track — signature pending, see §10 Q1). NEXT_AGENT_HANDOFF.md / FRESH_INSTANCE_HANDOFF.md are g6e-era history; live resume is PERSIST_RESUME.md → HANDOFF.md.

Completion raw JSONs (all committed, all seeds/pins inside): reports/m2c/arm_e_mondo__{0005180,0004975}.json (epoch summaries), reports/m2c/ot_panels/score_vectors/mondo__{0005180,0004975}__epoch_001.scores.json (pinned-order score vectors, order asserted), reports/m2c/loci_headtohead_mondo__{0005180,0004975}.json + reports/m2c/loci_headtohead_mondo__{0005180,0004975}_testknown.json (+ manifests — the §3.8 tables), reports/m2c/prospective_{null_pd,null_ad,model_pd,model_ad}.json, reports/m2c/reweighted_null_{pd,ad}.json, reports/m2c/gwas_loci_mondo__{0005180,0004975}.json, reports/m2c/vep_culprits_mondo__{0005180,0004975}.json, reports/m2c/ot_panels/ (panels + manifest + shas) + circularity_audit.json, reports/m2c/panels_wave2/ (wave-2 panels + manifest), reports/m4b/tabular_baseline{,_v1,_v2,_v3}_results.json, m5b/reports/{v5_results.json,hpo_results.json,verdict.md}. Off-box data copy: GitHub release data-m2c-2026-09-05 (data/processed/m2c tar.gz 58MB); GWAS bulk (707MB) + VEP cache are re-downloadable per PERSIST_RESUME.md Scenario C.

Checkpoints (M2b gitignored/EBS-only; M2c tracked in git per owner): M2b: reports/m2b/checkpoints_wave1/ (13 primaries ×3 seeds ≈39 88-105MB .pth), checkpoints_wave2_secondary/ (16), checkpoints_shuffle_controls/ (6), checkpoints_step0b/ (12) — SHA in funnel_provenance.json / wave*_manifest.json; off-host sina_full_backup_v2_2026-08-27.tar.gz 5.5GB. M2c (in-repo, each ~5MB): reports/m2c/checkpoints_completion_noana/mondo__{0005180,0004975}/{epoch_001,best,last}.pth + per-epoch JSONL; halted ANA wave reports/m2c/checkpoints_completion_ana/ preserved read-only (never resume per owner).

Repro (commands in §7): sina env pytest + ruff + report_lint (counts of record in reports/REPRODUCTION_2026.md), persist_restore.sh on fresh g6e.2xlarge (M2b) / PERSIST_RESUME.md stop→start on g7e (M2c).

---

7. How to Reproduce — for a fresh GPU host

```
# 0. Unpack v2 tarball (operator has it locally at ~/Downloads/
        sina_persist_backup.tar.gz)
# scp -i ~/.ssh/run3.pem ~/Downloads/sina_persist_backup.tar.gz
        ubuntu@54.202.31.103:/home/ubuntu/persist.tar.gz
sudo mkdir -p /opt/sina && sudo chown ubuntu:ubuntu /opt/sina
mkdir -p ~/restore && tar -xzf ~/
        sina_full_backup_v2_2026-08-27.tar.gz -C ~/restore
```

```
mv ~/restore/persist ~/persist && mv ~/restore/sina-platform ~/
        sina-platform
mkdir -p ~/miniconda3/envs && mv ~/restore/miniconda3/envs/sina ~/
        restore/miniconda3/envs/biopathnet ~/miniconda3/envs/
wget -q https://repo.anaconda.com/miniconda/Miniconda3-latest-
        Linux-x86_64.sh -O /tmp/mc.sh && bash /tmp/mc.sh -b -p ~/
        miniconda3
cd ~/sina-platform && bash scripts/pod/persist_restore.sh   #
        rebuilds /opt/sina

# 1. Verify restore
~/miniconda3/bin/conda run -n sina python -m pytest tests/unit -q
ls reports/m2b/checkpoints_wave1 | wc
        -l                                  # 13+ (39 with 3 seeds)
ls /opt/sina/data/processed/m2/lodo_exports | wc
        -l                  # 29
~/miniconda3/bin/conda run -n sina python -m sina.lint_reports
        reports/ --config configs/report_lint.yaml  # 0 violations

# 2. Rerun any piece
git checkout 53f24ca
conda run -n sina python -m sina.lint_reports reports/ --config
        configs/report_lint.yaml
conda run -n sina python -m pytest tests/unit -q
python scripts/m2_build_lodo_exports.py --config configs/
        m2_lodo_exports_keep.yaml --resume
P=~/miniconda3/envs/biopathnet; OV=~/persist/cuda-11.8-overlay;
        PATH="$OV/hostbin:$PATH" CUDA_HOME="$OV" CXX="$P/bin/
        x86_64-conda-linux-gnu-g++" ~/miniconda3/bin/conda run -n
        biopathnet python -u scripts/pod/gate0_ladder.py --config
        configs/m2b_gate0_ladder.yaml
P=~/miniconda3/envs/biopathnet; OV=~/persist/cuda-11.8-overlay;
        PATH="$OV/hostbin:$PATH" CUDA_HOME="$OV" CXX="$P/bin/
        x86_64-conda-linux-gnu-g++" ~/miniconda3/bin/conda run -n
        biopathnet python -u scripts/pod/run_wave1_benchmark.py --
        batch-size 32 --seed 1337
conda run -n sina python scripts/m2_wave_stats.py --config
        configs/m2_wave_stats.yaml
python scripts/pod/extract_paths.py --config configs/
        m3_pathway_funnel.yaml --disease mondo:0010196
python scripts/pod/anchored_topology_null.py --config configs/
        m3_pathway_funnel.yaml
```

Batch 32 pinned (29.28GB on L40S). RSS <40GB. no NeighborLoader. CUDA 11.8 overlay required (host gcc-15 incompatible). See FRESH_INSTANCE_HANDOFF.md and ~/persist/RECOVERY.md for full detail.

```
# 3. Rerun any completion-track piece (g7e Blackwell; GPU train
        needs the §1 env
#    line from HANDOFF.md — overlay hostbin + bpn120; every eval
        below is CPU-only)
P=~/miniconda3/envs/bpn120; OV=~/persist/cuda-12.8-overlay
export PATH="$OV/hostbin:$P/bin:$PATH" CUDA_HOME=$P
        TORCH_EXTENSIONS_DIR=/tmp/torch_ext_probe
CUDA_VISIBLE_DEVICES="" $P/bin/python -u scripts/pod/
        m2c_pd_test_once.py --unit mondo__0004975 \
```

```
  --exports-dir data/processed/m2c/completion_exports \
  --ckpt-root reports/m2c/checkpoints_completion_noana --
        checkpoints epoch_001.pth \
  --threads 3 --out /tmp/ad_test_once.json   # MRR 0.01061 H@10
        0.01610
CUDA_VISIBLE_DEVICES="" $P/bin/python -u scripts/pod/
        m2c_ot_panel_eval.py score \
  --unit mondo__0004975 --exports-dir data/processed/m2c/
        completion_exports \
  --checkpoints reports/m2c/checkpoints_completion_noana/
        mondo__0004975/epoch_001.pth \
  --dump-scores-dir reports/m2c/ot_panels/score_vectors --threads
        3 --out /tmp/ad_ot.json
~/miniconda3/envs/sina/bin/python scripts/pod/
        m2c_prospective_model.py --unit mondo__0004975 \
  --scores reports/m2c/ot_panels/score_vectors/
        mondo__0004975__epoch_001.scores.json \
  --loci-json reports/m2c/gwas_loci_mondo__0004975.json \
  --exports-dir data/processed/m2c/completion_exports --out /tmp/
        probe_ad.json
~/miniconda3/envs/sina/bin/python scripts/pod/
        m2c_reweighted_null.py --unit mondo__0004975 \
  --exports-dir data/processed/m2c/completion_exports --out /tmp/
        rew_ad.json   # gate re-check, seconds
# 3b. Head-to-head (CPU, sina env; needs completion exports + OT
        cache + score vectors)
~/miniconda3/envs/sina/bin/python scripts/pod/
        m2c_loci_headtohead.py gate --unit mondo__0005180   #
        reproduces 0.904
~/miniconda3/envs/sina/bin/python scripts/pod/
        m2c_loci_headtohead.py run --unit mondo__0005180 \
  --known-split test --out reports/m2c/
        loci_headtohead_mondo__0005180_testknown.json \
  --report reports/m2c/loci_headtohead_mondo__0005180_testknown.md
# (repeat gate+run for mondo__0004975; default --known-split
        train2 regenerates the full panel)
# 4. Tabular/functional controls (CPU, sina env)
conda run -n sina python scripts/m4b/tabular_baseline.py report --
        featureset v3   # byte-identical regen from data/
        processed/m4b/train_results_v3.json
# M5B verdict numbers derive from m5b/reports/v5_results.json (see
        reports/REPRODUCTION_2026.md item 8)
ruff check scripts/ src/ tests/   # 0
PYTHONPATH=src python -m sina.lint_reports reports/ --config
        configs/report_lint.yaml   # 0
PYTHONPATH=src:. python -m pytest tests/unit -q
```

Resume after stop→start is `PERSIST_RESUME.md` (start box → `git pull` → live-jobs section); the EBS root persists the repo, both conda envs, and `~/persist`. `NEXT_AGENT_HANDOFF.md` is pre-Blackwell history — do not follow its env pins on `sm_120`.

8. References

Menche et al. Nature 2015 (disease modules); Himmelstein et al. eLife 2017 (Hetionet DWPC); Subramanian et al. PNAS 2005 (GSEA); Barabási et al.; Oleshko et al. bioRxiv 2024 BioKGC; Hu et al. Nat Biomed Eng 2026 BioPathNet 10.1038/s41551-025-01598-z; Huang et al. Nat Med 2024 TxGNN Explainer; PaGE-Link WWW 2023; PoPS Nat Genet 2023; GNNExplainer Ying et al. NeurIPS 2019; GNNenrich; Jain & Wallace NAACL 2019 Attention is not Explanation; Sundararajan et al. 2017 Integrated Gradients; López-Otín et al. Cell 2013/2023 Hallmarks of Aging; OptimusKG 10.7910/DVN/IYNGEV; PrimeKG 10.7910/DVN/IXA7BM; MONDO CC-BY 4.0; GWAS Catalog (Sollis et al. NAR 2023, EBI FTP bulk); Ensembl VEP (McLaren et al. Genome Biol 2016, REST); Open Targets Platform + Genetics (Ghoussaini et al. NAR 2021, association panels + gold-standards); Schipper, de Leeuw, Posthuma et al. (FLAMES, VU Amsterdam); Mountjoy et al. Nat Genet 2021 (OT L2G).

---

9. Appendices

A. Registry excerpt (`data/raw/REGISTRY.json` 49 assets)

mondo_obo 2026-07-24 53,086,111 bytes sha256 fce4bccd..., optimuskg 37 files 21,813,816 edges, primekg 8,100,498, txgnn neurodigenerative 68 diseases, anyburl 23-1x BSD-3. Full SHA-256 per asset in REGISTRY.json.

B. Config hashes (pinned)

m2_trackd_relations.yaml:status APPROVED 2026-08-15, m2_lodo_exports_keep.yaml, m2_lodo_shuffles_keep.yaml, m2_lodo_baselines_keep.yaml, m2_wave_stats.yaml (Wilcoxon 2^13 + BCa 1000), m3_pathway_funnel.yaml (top_k 100 [50,100,200], path_topk 10, max_hops 6, grad×PNA, min_path_score 0.01, min_support 5, log(degree)), m2b_gate0_ladder.yaml.

C. Per-unit train2 / valid / test (from `lodo_exports_keep.md`)

Full 29 rows in reports/m2b/lodo_exports_keep.md; train2 52-730 (primaries), valid 7-126, test 52-730, ctx exp = ctx aud (all retained), anc 9-45, sib 5-185.

D. Encoding worked example (for the judge who wants jargon + meaning)

Query: `Werner --ASSOCIATED_WITH--> ?` with path `Werner --DIS-PHE--> HP:0004322 --PHE-GEN--> WRN`. * `query vector = Embedding(ASSOCIATED_WITH) 32-dim` (learned table `54×32`). `Werner_t0 = query`, others `0` (`boundary` in `model.py:120`). * `relation vectors = Embedding(27 relations, 32)` + `Linear` (`layer.py:53,55`) — `DIS-PHE`, `PHE-GEN`, `GEN-GEN` etc start uniform random, end learned via `Adam`. * `msg = node_state * relation_vector` (`distmult`), `aggregate = PNA(mean,max,min,std → Linear)`. After 2 layers, `WRN_t2 = (Werner * DIS-PHE) * PHE-GEN`. Score = `MLP( WRN_t6 concat query )`. Both SINA and BioPathNet paper use identical `32×6 distmult/pna` — failure is task difficulty (LODO hide whole disease vs random hide 10% edge), not encoding.

---

10. Peer-review brief — project state, open questions, how to give feedback

Read this section if you are a reviewer (human or agent) asked to critique SINA. Everything below resolves to files listed in §6; no conclusions are sought beyond ideas — binding decisions stay with the owner.

One-paragraph state (2026-10-06): SINA benchmarks GNN disease-gene recovery on OptimusKG two ways — LODO zero-shot transfer (M2b: `mean MRR 0.00775`, not significant, leakage-certified by shuffle collapse) and seen-disease completion (M2c: PD `0.01887/0.02950`, AD `0.01061/0.01610`, stack-invariant plateaus). The September product hypothesis — per-locus known-recovery at `0.904/0.954` top-1 — is now falsified by its own leakage-controlled split (§3.8): those knowns are train positives, and on genes never seen in training the model loses to (PD `0.300` vs `0.600`) or ties (AD `0.1613` vs `0.2258`) Open Targets L2G, below chance-draw means. Two CPU controls close the loop: a fresh-feature tabular model ties OT (`+0.006`, §3.9) and five functional-evidence tracks all lose (best `0.300` vs `0.477`, §3.10). The program's remaining value is the certified negative plus the harness that produced it. Box `g7e` is stopped with zero live jobs; resume is `PERSIST_RESUME.md`.

What changed since the 2026-09-06 snapshot (the previous version of this section): the head-to-head vs OT L2G landed (`2026-09-11`, train-knowns WIN `+0.29/+0.42`); the test-knowns re-grade inverted it (`2026-09-25`, PD LOSS / AD TIE, §3.8); a committed-renderer bug that showed full-panel numbers inside the test-knowns reports was found and fixed (`12898d5`, `2026-10-06` — the JSONs were always correct; the `.md` text was wrong); M4B v1→v4 completed (covered-subset TIE vs OT, first significant win over nearest-gene, FLAMES correction, fresh-feature v4 LOSS,

§3.9); M5B completed (five functional tracks, LOSS on all arms, §3.10). Open review Q4 from 09-03 (“is ANY KG claim viable while the gap overlaps the null?”) is answered in the negative for genome-wide discovery and for unseen-gene per-locus recovery.

Open questions for feedback (ordered by decision value): 1. `CHANGES/0027` is still unsigned — does the AD replication (`§3.7`) satisfy you as the second-disease invariance evidence for signing rank+noANA as the criterion, given AD’s `CI`s overlap floor/null? 2. ~~Per-locus recovery beats degree by `+0.29` (PD) / `+0.14` (AD) — enough to justify building the “rank the knowns” workflow?~~ **ANSWERED 2026-09-25 (§3.8):** no. The margin was train-positive memorization; test-knowns show no unseen-gene signal. Workflow hypothesis closed. 3. The power rule (`n≥15` novel functional singletons) killed the final test at `n=2`. Is QTL-colocalization the cheapest credible next gold, or is the certified negative sufficient for the preprint as framed? 4. Any leakage vector we missed? Candidates: `PHE-GEN` edges carrying disease signal (`pheno_shared` is the best independent generator — coincidence or leakage?), `DIS-DIS` hierarchy edges, train2 co-count structure the null already exploits. 5. Is the document-only default on the aux misroute (§2.9) defensible, or does the accidental `GO-CC + HPO` aux mix undermine the plateau reading enough to require the fixed-aux rerun? 6. Given §3.8-3.10, is there any remaining cheap experiment that would change the framing of the negative result — or is the honest move the preprint as written?

How to give feedback: cite file + line/number, propose the concrete next experiment with its falsification bar (form: “if X < Y, kill Z”), and flag any number in §3 that does not reproduce from its raw JSON in §6 via the §7 commands. Do not relaunch GPU training, do not touch `checkpoints_completion_ana/` (read-only audit trail), do not score hidden tests except through the committed scripts.

This paper is the single record on *`main`* (self-contained). To re-verify: *`conda run -n sina python -m sina.lint_reports reports/ --config configs/report_lint.yaml`* and *`conda run -n sina python -m pytest tests/unit -q`*; clean-machine verification log: *`reports/REPRODUCTION_2026.md`*.